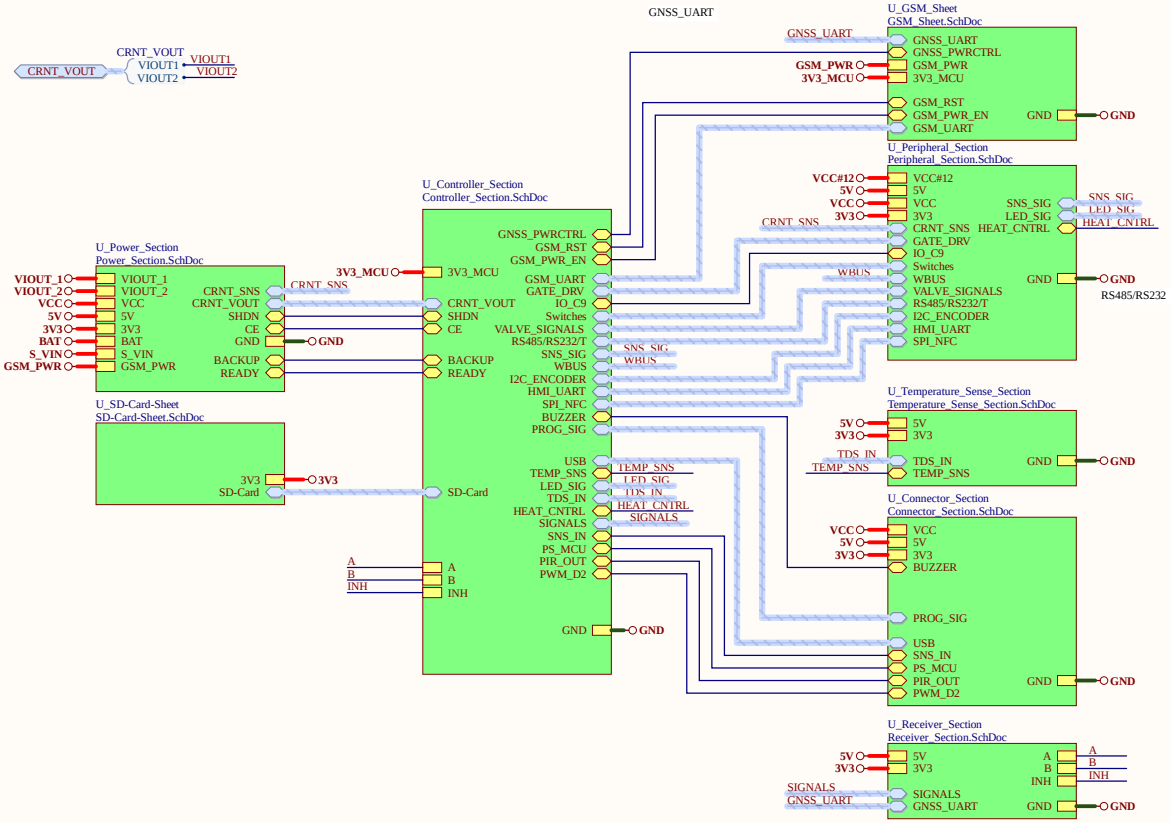
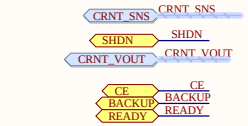
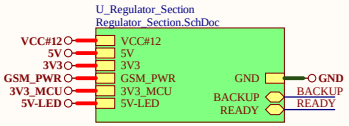


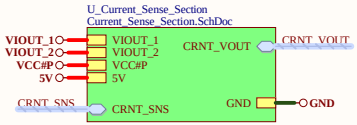
Title: Top_Sheet.SchDoc

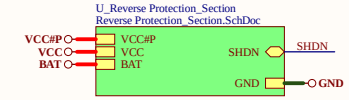
File: GTV3.4_Board.PrjPcb

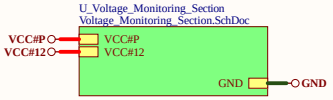
Date: 11/19/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 17

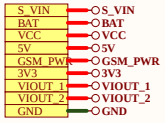


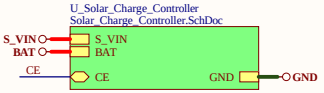








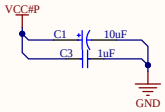




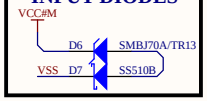
eSpark Consulting Group

Title: Power_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 11/19/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 17

DECOUPLING CAPACITOR

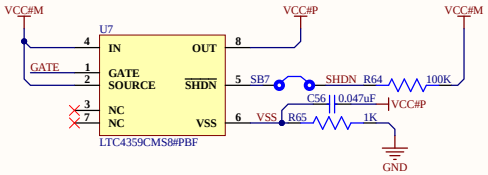


INPUT DIODES

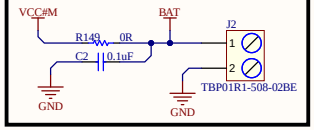




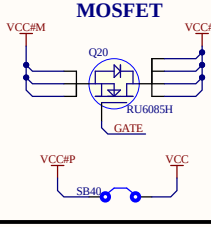
REVERSE POLARITY PROTECTION IC



BATTERY / MAIN INPUT



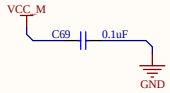
MOSFET



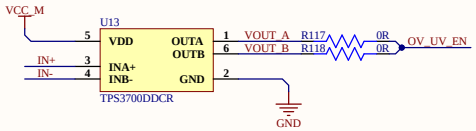
eSpark Consulting Group

Title: Reverse Protection_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 17

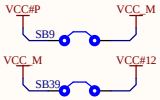
DECOUPLING CAPACITOR



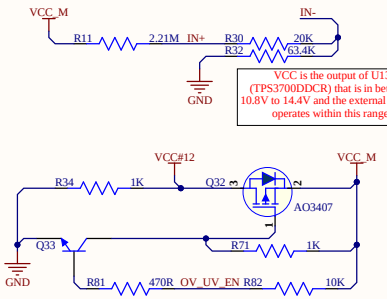
UNDER AND OVER VOLTAGE PROTECTION IC

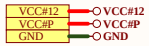


VOLTAGE TO MONITOR



SUPPORTING CIRCUITRY

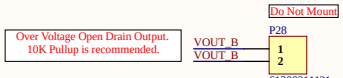




UNDER VOLTAGE OUTPUT

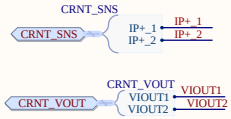


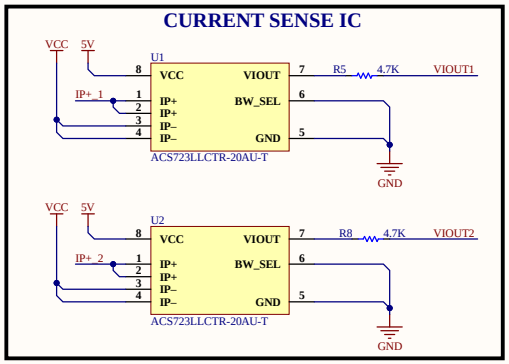
OVER VOLTAGE OUTPUT



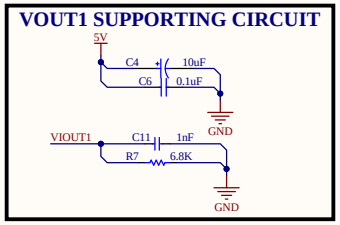
eSpark Consulting Group

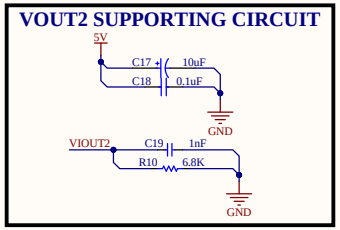
Title: Voltage_Monitoring_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 17

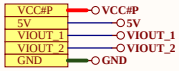




Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.

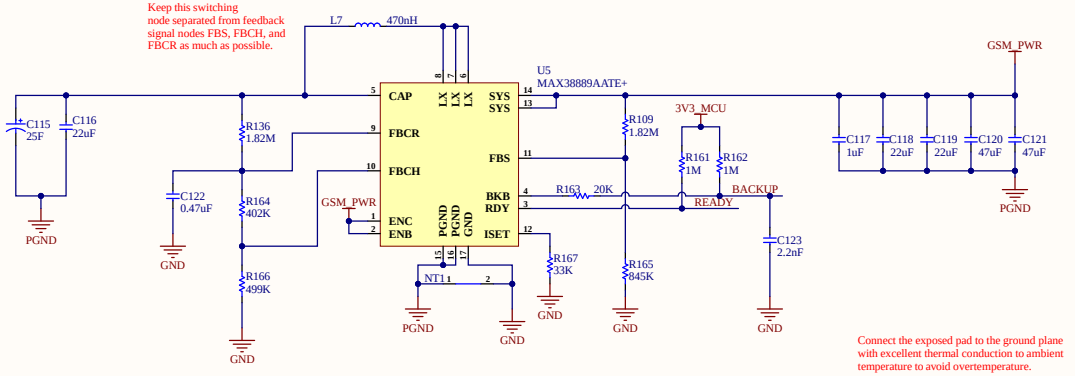




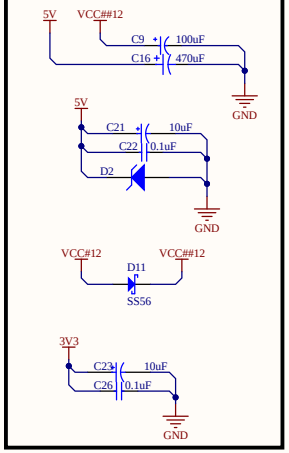


Title: Current_Sense_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 17

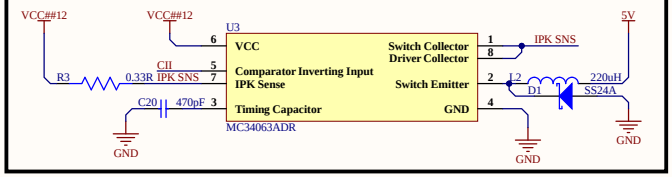
SUPERCAPACITOR BACKUP CONTROLLER



DECOUPLING CAPACITOR

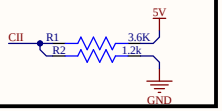


5V BUCK REGULATOR

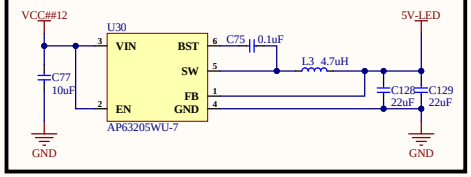


VOUT = 1.25 (1+ R2/R1)
R2=3.6K
R1=1.2K

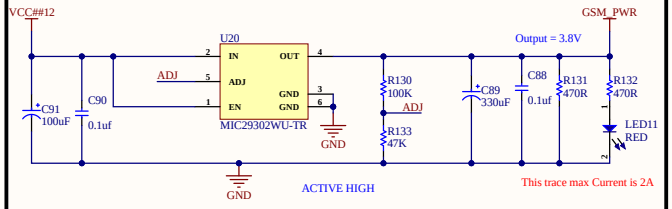
SUPPORTING CIRCUIT

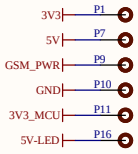


RGB LED BUCK

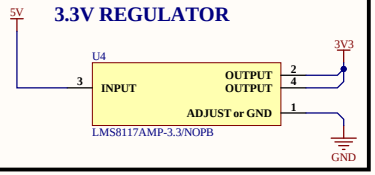


REGULATOR FOR GSM POWER

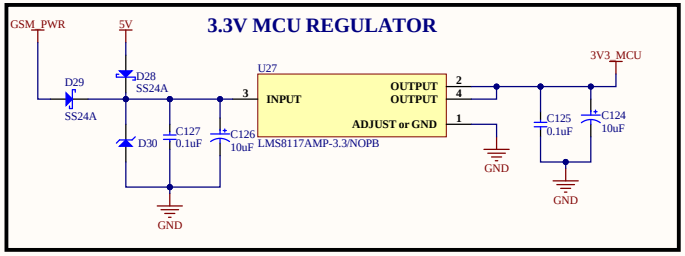




3.3V REGULATOR



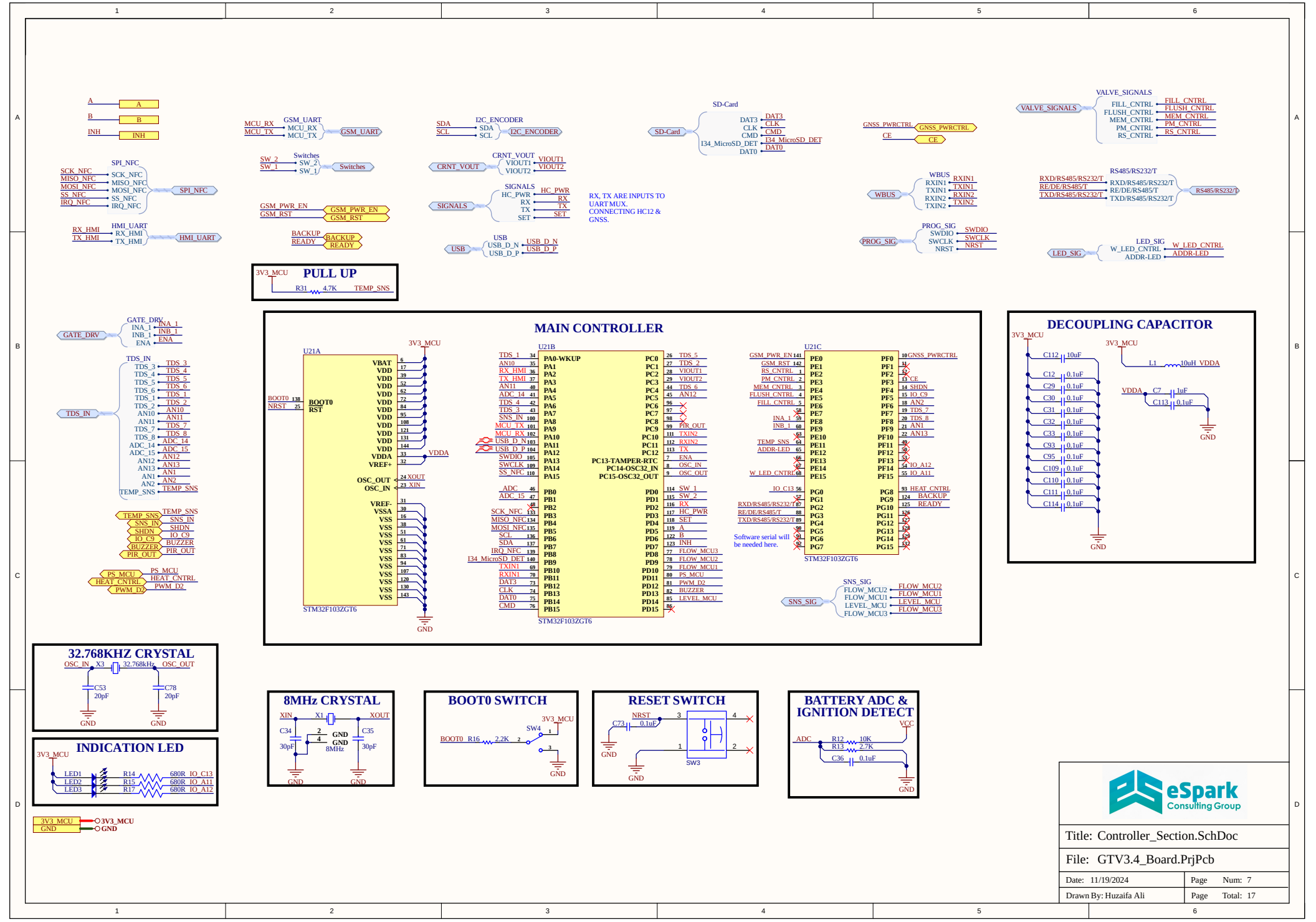
3.3V MCU REGULATOR



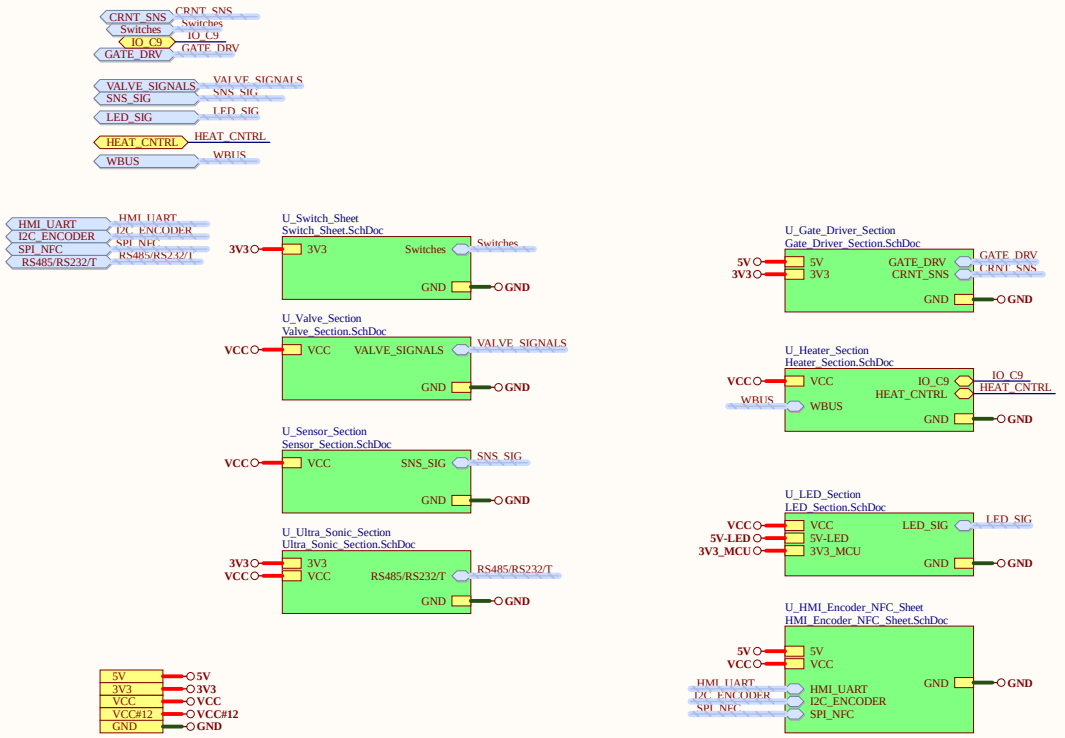
5V-LED	5V-LED
VCC#12	VCC#12
5V	5V
GSM_PWR	GSM_PWR
3V3	3V3
3V3_MCU	3V3_MCU
GND	GND
BACKUP	BACKUP
READY	READY

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Title: Regulator_Section.SchDoc		
File: GTV3.4_Board.PrjPcb		
Date: 11/20/2024	Page	Num: 6
Drawn By: Huzaifa Ali	Page	Total: 17

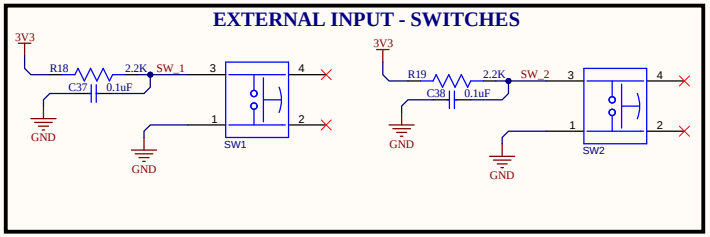


Title:	Controller_Section.SchDoc
File:	GTV3.4_Board.PrjPcb
Date: 11/19/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 17



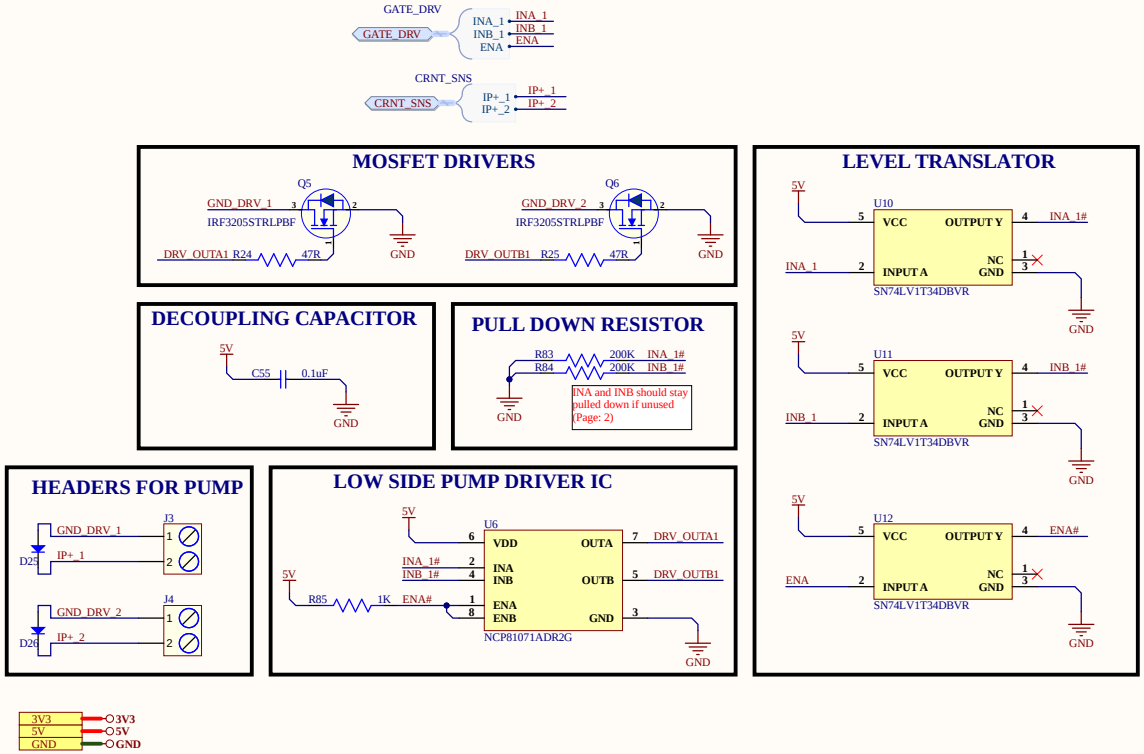
Title: Peripheral_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 11/19/2024	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 17





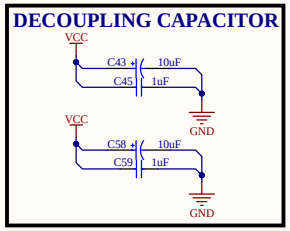


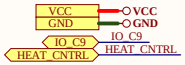
Title: Switch_Sheet.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 17



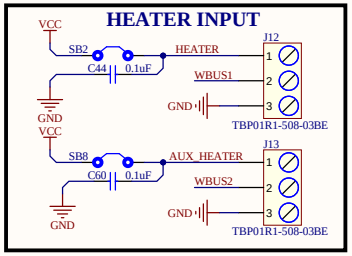
eSpark Consulting Group

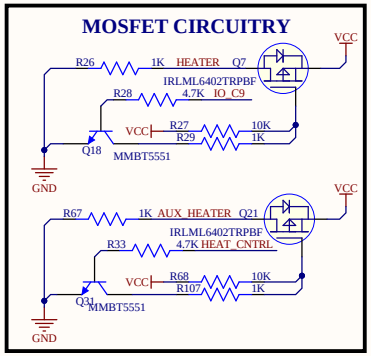
Title: Gate_Driver_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 17

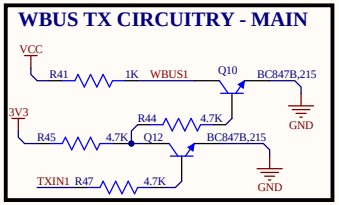


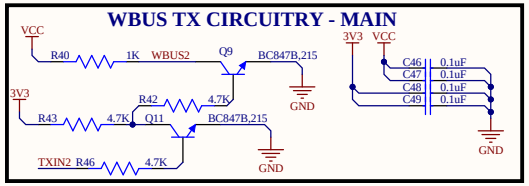


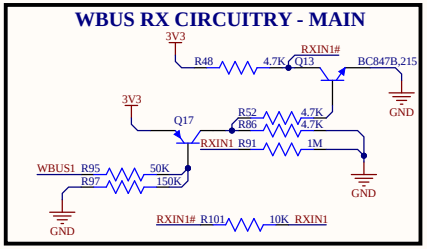


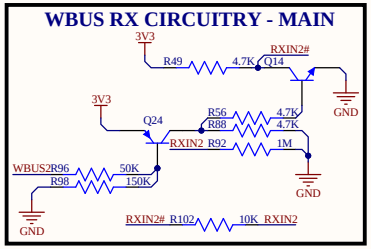


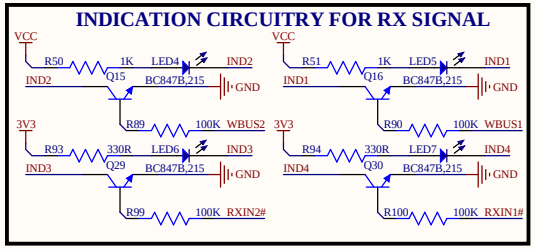








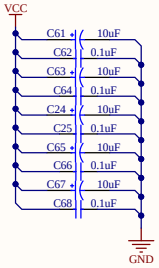




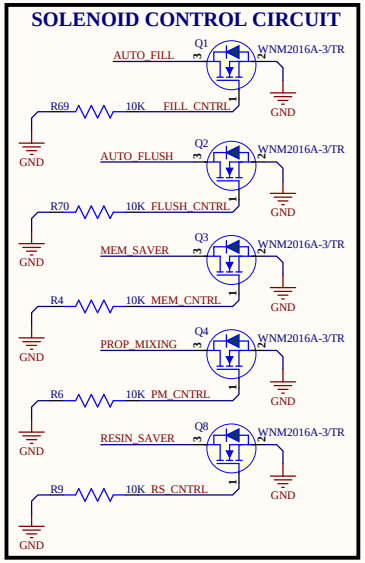
eSpark Consulting Group

Title: Heater_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 11
Drawn By: Huzaifa Ali	Page Total: 17

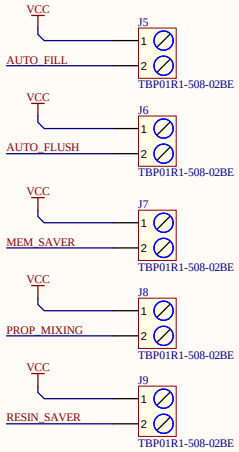
DECOUPLING CAPACITOR



SOLENOID CONTROL CIRCUIT



SOLENOID VALVE INPUTS



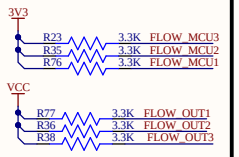




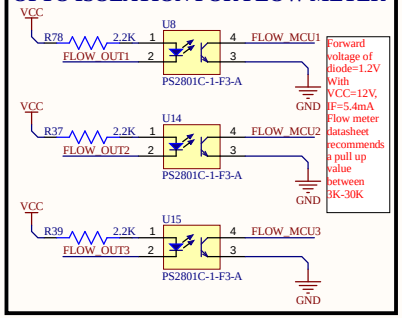
eSpark Consulting Group

Title: Valve_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 12
Drawn By: Huzaifa Ali	Page Total: 17

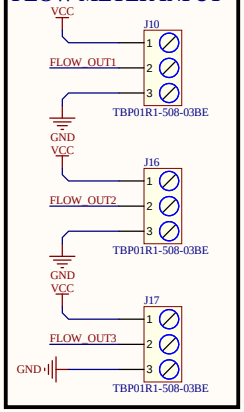
PULL UP RESISTORS



OPTO ISOLATION FOR FLOW METER



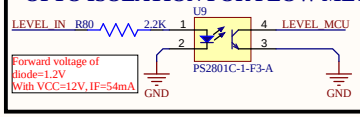
FLOW METER INPUT



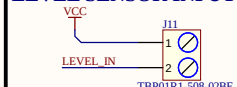
SUPPORTING CIRCUIT

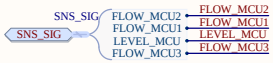


OPTO ISOLATION FOR FLOW METER



LEVEL SENSOR INPUT



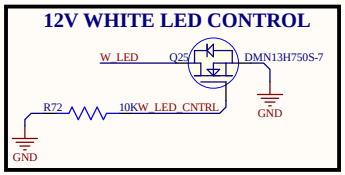


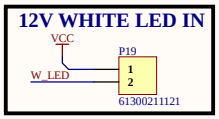


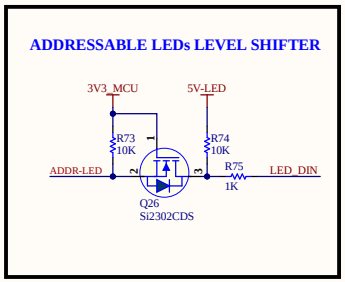
eSpark Consulting Group

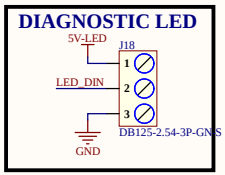
Title: Sensor_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 13
Drawn By: Huzaifa Ali	Page Total: 17

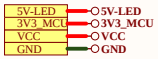








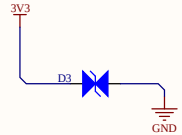




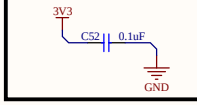
eSpark Consulting Group

Title: LED_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 11/19/2024	Page Num: 14
Drawn By: Huzaifa Ali	Page Total: 17

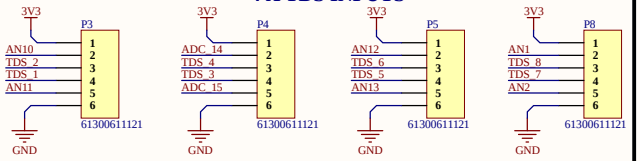
PULL UP RESISTORS



DECOUPLING CAP

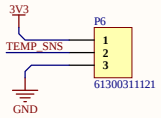


4 X TDS INPUTS

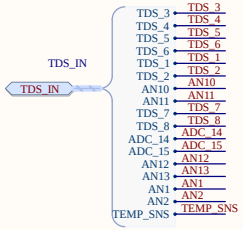


These TDS headers connect with TDS header on V3 Controller PCB via Female to Female Cable.

TEMP SENSOR DS18B20



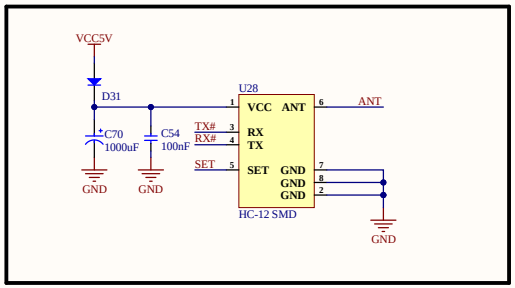




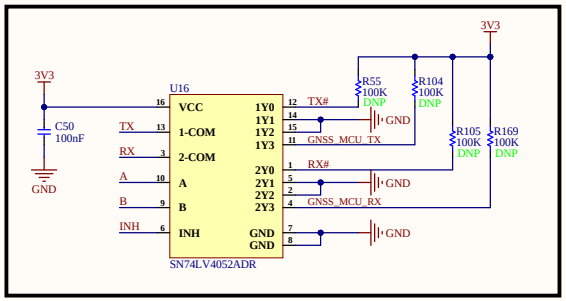
eSpark Consulting Group

Title: Temperature_Sense_Section.SchDoc	
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Date: 3/7/2024	Page Num: 15
Drawn By: Huzaifa Ali	Page Total: 17

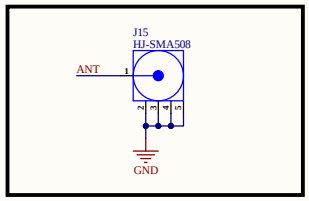
HC-12 433MHz MODULE



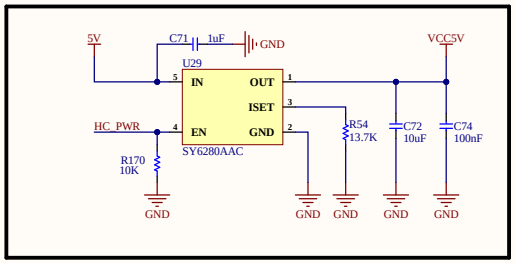
UART MUX

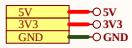


SMA CONNECTOR

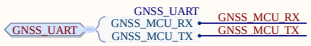


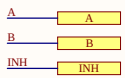
POWER SWITCH





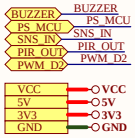




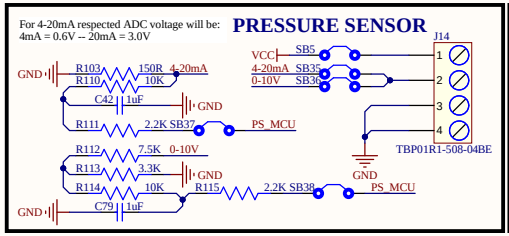


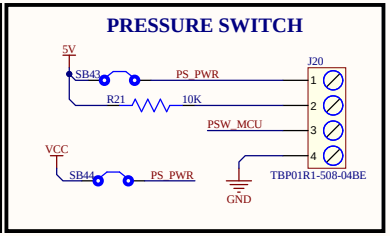
eSpark Consulting Group

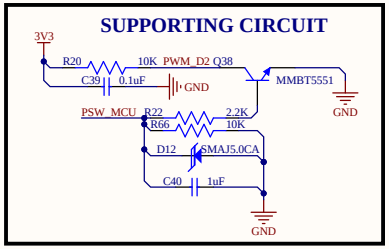
Title: Receiver_Section.SchDoc		
File: GTV3.4_Board.PrjPcb		
Date: 11/19/2024	Page	Num: 16
Drawn By: Huzaifa Ali	Page	Total: 17

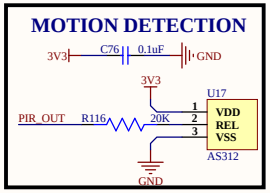


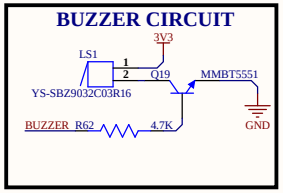


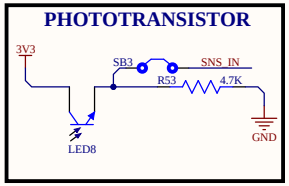


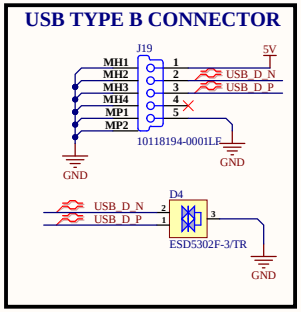


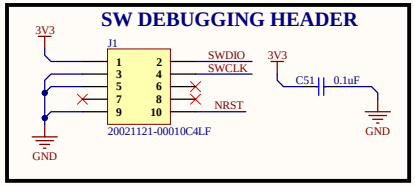






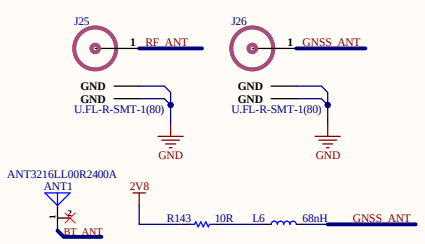




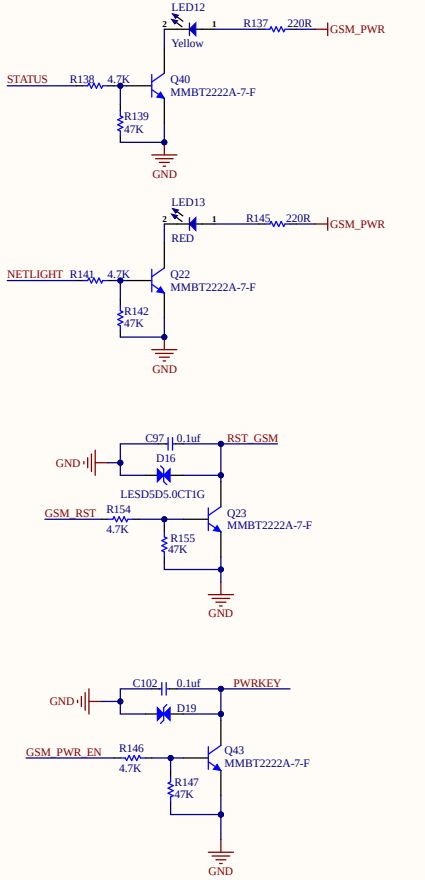


Title: Connector_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 17
Drawn By: Huzaifa Ali	Page Total: 17

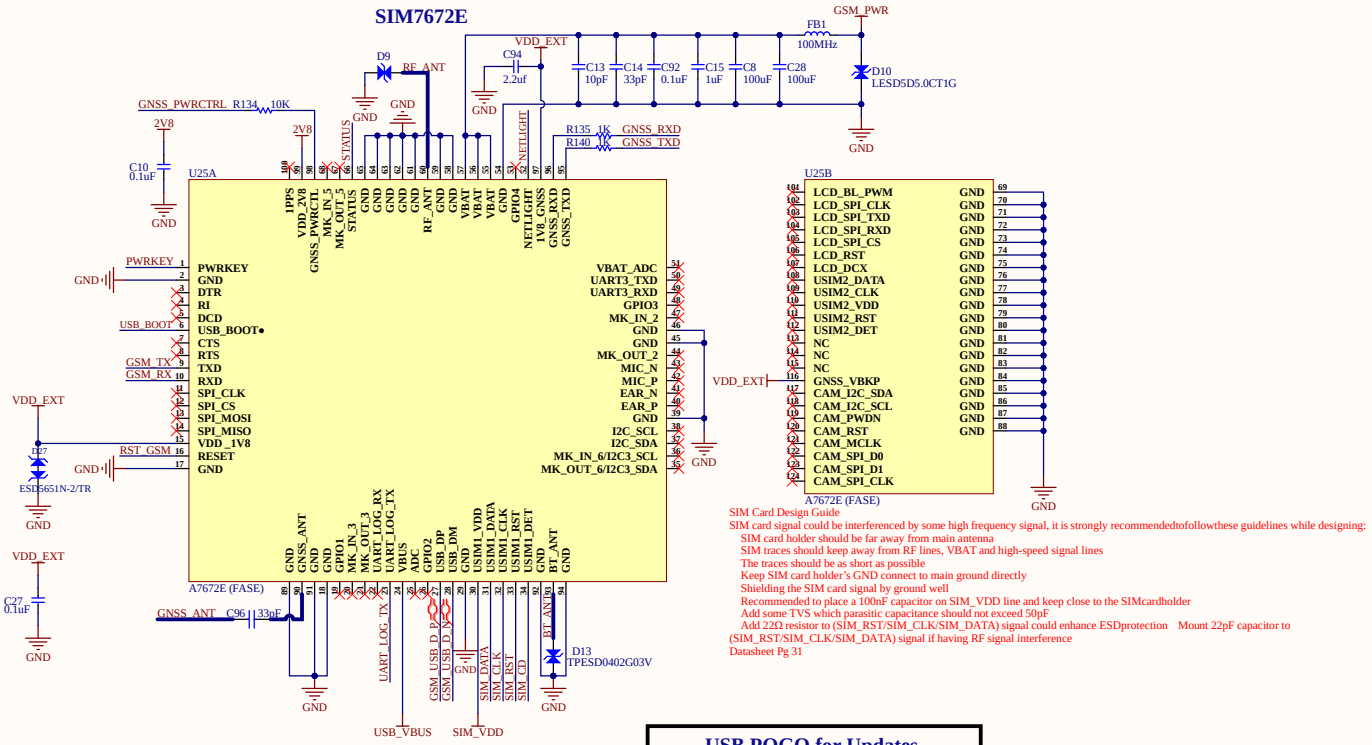
ANTENNA



GSM_COMMANDS



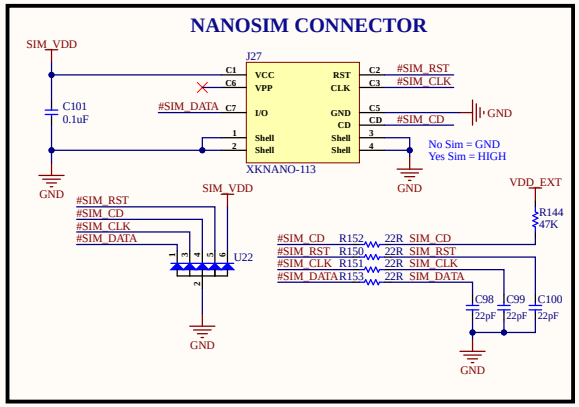
SIM7672E



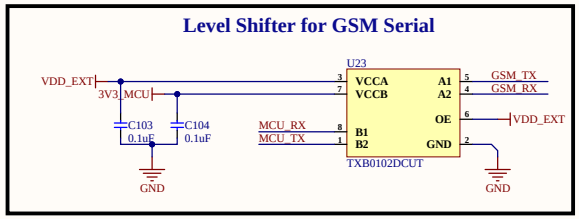
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

Datasheet Pg 31

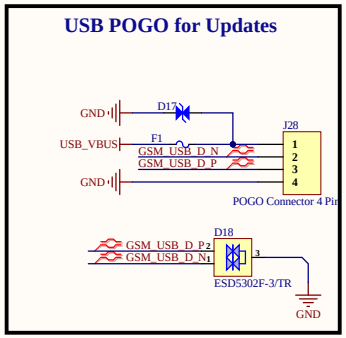
NANOSIM CONNECTOR



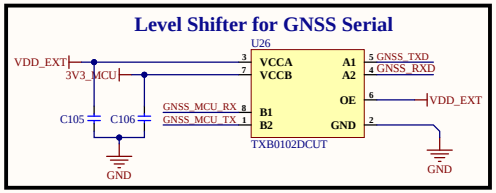
Level Shifter for GSM Serial

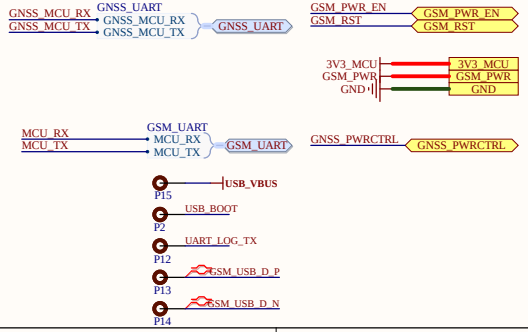


USB POGO for Updates



Level Shifter for GNSS Serial

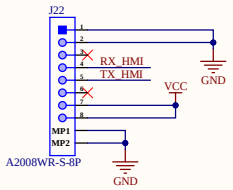




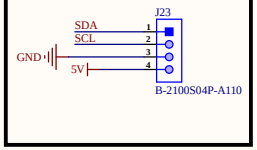
eSpark Consulting Group

Title: GSM_Sheet.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *

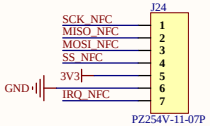
HMI INPUT STONE



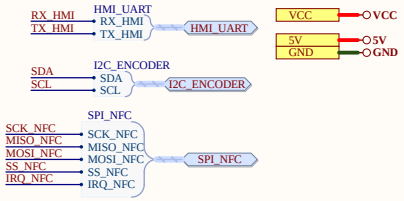
ENCODER INPUT



NFC INPUT

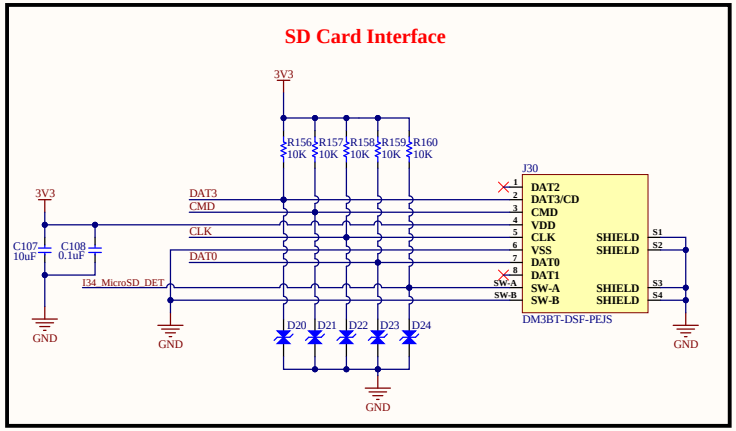


PN532 NFC Module 13.56Mhz



eSpark Consulting Group

Title: HMI_Encoder_NFC_Sheet.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *



SD-Card

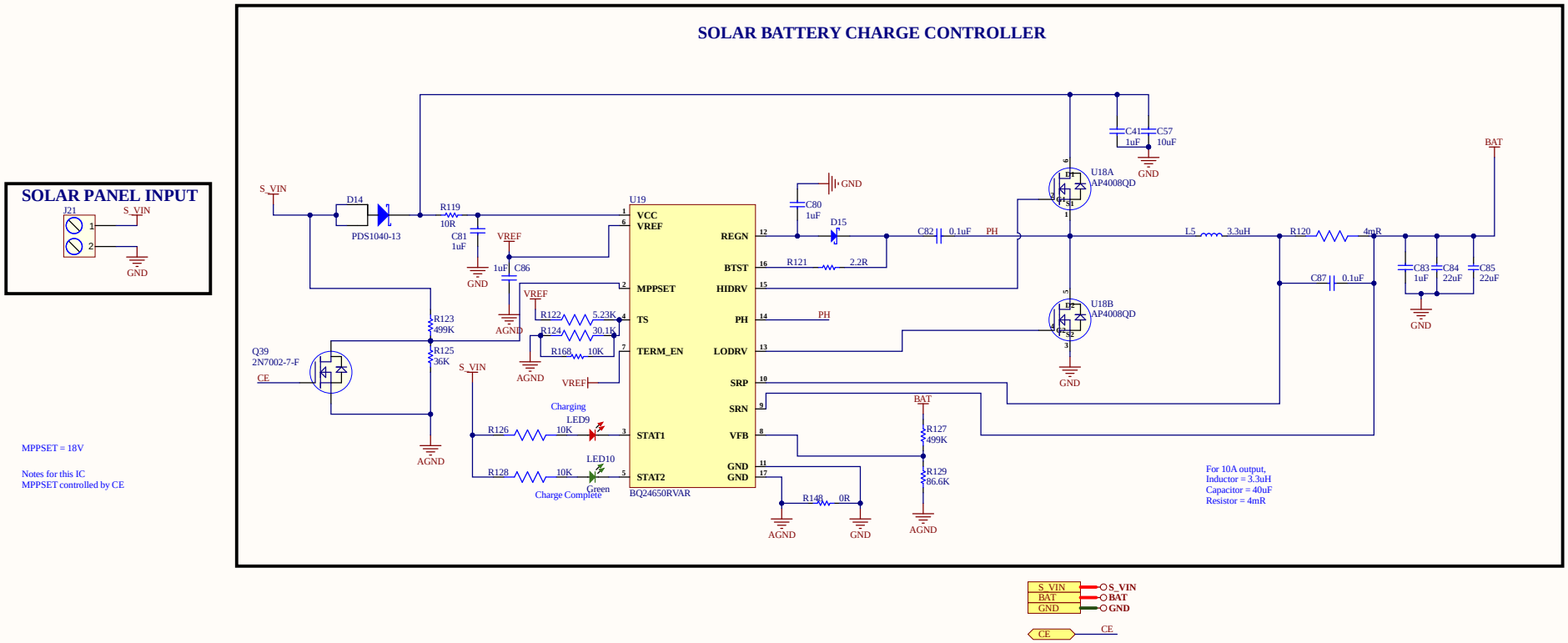
SD-Card

DAT3 DAT3
CLK CLK
CMD CMD
I34_MicroSD_DET I34_MicroSD_DET
DAT0 DAT0

3V3 3V3

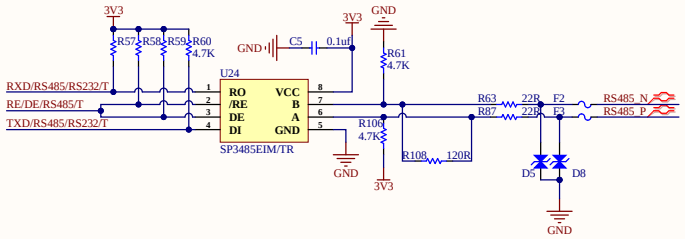
eSpark Consulting Group

Title: SD-Card-Sheet.SchDoc	
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Date: 3/7/2024	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *

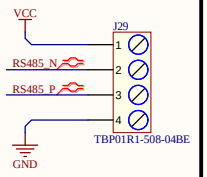


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Date: 3/7/2024	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *

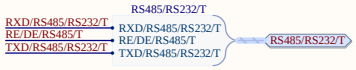
RS485 to UART



ULTRA-SONIC SENSOR



Part Selected:
https://www.massa.com/industrial/ultrasonic-sensors/#1501873783525-5657d00f-07c9





eSpark Consulting Group

Title: Ultra_Sonic_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *